Masked Linear Cross-Attention for Robust RGB-D Semantic Segmentation under Incomplete Depth

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1. INTRODUCTION

Indoor robots, AR devices and mapping systems depend on RGB-D sensors, yet active depth sensing fails in systematic ways: reflective, transparent, dark or distant surfaces leave holes, and edges or multipath effects produce noisy values. Most RGB-D semantic segmentation methods fuse every depth position as if it were valid, so missing measurements, usually stored as zeros, propagate into the fused features. Cross-modal attention, as in CMX [1], improves fusion but has quadratic cost and no explicit notion of measurement validity.

We ask: *to what extent does excluding invalid depth positions inside a linear cross-attention reduce the accuracy loss of RGB-D segmentation when depth is noisy, partially missing or absent, at a small computational cost?* We do not propose new encoders; we isolate the effect of validity masking inside linear attention and will report it whether it helps or not. Our contributions are (C1) MLCA, a lightweight masked linear cross-attention for RGB and depth fusion; (C2) controlled ablations under partial, contiguous, noisy and total depth loss; and (C3) an accuracy–robustness–cost analysis.

2. METHODOLOGY

Architecture (Fig. 1). RGB features come from a frozen EfficientViT-B1 [2], whose lightweight multi-scale linear attention targets high-resolution dense prediction. Depth features come from a frozen DeFM ResNet-18 [3], a CNN distilled from a self-supervised foundation model trained only on depth images, whose log-compressed input encoding keeps metric-scale information. After spatial alignment and projection to a common width, MLCA uses RGB tokens as queries and depth tokens as keys and values, with kernelized linear attention [4]:

$$MLCA(Q,K,V)_i = \varphi(Q_i)^\top \Sigma_j\, m_j\, \varphi(K_j)\, V_j \;/\; (\, \varphi(Q_i)^\top \Sigma_j\, m_j\, \varphi(K_j) + \varepsilon\,)$$

where $m_j \in \{0,1\}$ marks depth token j as valid when the fraction of valid pixels in its patch exceeds a threshold. Masked tokens are removed from both sums, so the cost stays linear in the number of tokens, and the output vanishes when no depth is valid. The MLCA output is added to the RGB features through a residual path, so the model falls back to an RGB-only prediction when depth is absent. Only the projections, MLCA and a lightweight decoder are trained.

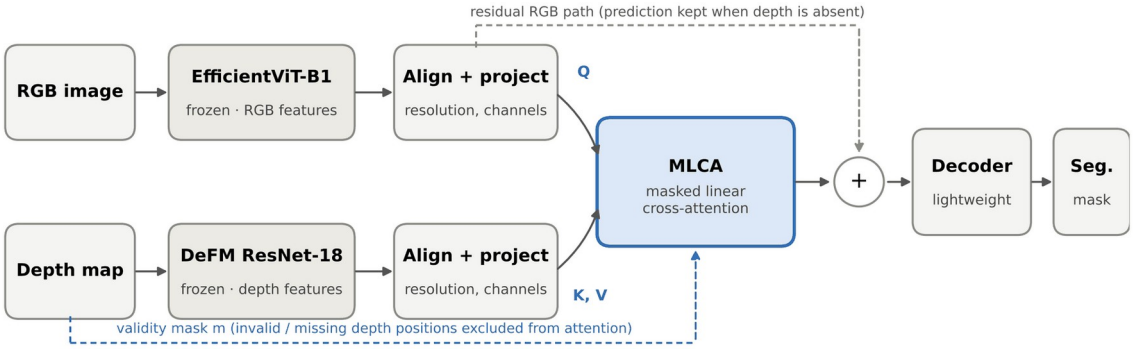


Fig. 1. Proposed architecture. Both encoders are frozen; the validity mask is computed from the raw depth map.

Data and protocol. Training, ablations and robustness tests use NYU Depth V2 [5] (1,449 labeled indoor RGB-D images, standard 795/654 split). Five configurations add one component at a time (Table 1). Each is evaluated with complete depth, random holes of 10%, 30% and 50%, contiguous missing regions closer to real sensor failures, Gaussian

noise, and total depth absence. We report mIoU, per-class IoU, mIoU drop relative to complete depth, parameters, FLOPs, latency and VRAM, as mean ± std over three seeds.

Table 1. Ablation design on NYU Depth V2. Each step adds one component.

ID	Configuration	Isolates the effect of
E1	EfficientViT-B1, RGB only	Baseline
E2	+ DeFM depth, concatenation	Adding depth
E3	+ linear cross-attention, no mask	Attention-based fusion
E4	+ validity mask (MLCA)	Masking invalid depth
E5	E4 with / without modality dropout	Architecture vs. training with failures

3. RESULTS AND DISCUSSION

Status. No experimental results are available at the time of submission. The architecture, degradation protocol and analysis plan are defined, and configurations E1–E5 are under way. The main evidence will be robustness curves (mIoU of E1–E5 versus the percentage of missing depth) and a cost–accuracy comparison across configurations. We state the expected outcome of each ablation before running it:

E2 vs. E1: depth should raise mIoU with complete depth but degrade it as holes grow, since zeros enter the fused features.

E3 vs. E2: attention should give better fusion with complete depth, with a similar sensitivity to missing depth.

E4 vs. E3 (main hypothesis): a smaller mIoU drop under random and contiguous holes, little difference with complete depth, and performance close to E1 when depth is absent, thanks to the residual path. We expect no gain under Gaussian noise, because a binary mask does not flag values that are present but wrong.

E5: modality dropout alone may yield similar robustness. If so, the benefit would come from training rather than from the architecture, and we will report it as a negative result for MLCA.

What would change our mind: if E4 ≈ E3 under holes, masking adds nothing; if E5 without the mask ≈ E4, robustness comes from training, not from the operator.

4. CONCLUSIONS

Expected main finding. MLCA is expected to reduce the mIoU drop relative to unmasked fusion when depth is incomplete or missing, and to stay close to RGB-only performance when depth is absent, with a bounded increase in parameters and latency.

Relevance. The study would quantify the specific contribution of validity masking in linear attention and support efficient RGB-D segmentation that is robust to sensor failures, useful for robotics and edge devices.

Limitations. The novelty is incremental, since the components already exist; a binary mask does not detect depth that is present but noisy; NYU Depth V2 is small; the feature extractors are frozen; and sensor failures are simulated.

Next steps. Implement and evaluate E1–E5 on NYU Depth V2 under all degradation conditions with three seeds, confirm novelty against the literature, and add a reliability estimate for noisy depth.

Code: https://github.com/joagovi/inf659-rgbd-mlca. **Acknowledgments:** Escuela de Posgrado PUCP, Vicerrectorado de Investigación PUCP and Grupo IA-PUCP.

REFERENCES

[1] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," IEEE Trans. Intell. Transp. Syst., vol. 24, no. 12, 2023.
[2] H. Cai, J. Li, M. Hu, C. Gan, and S. Han, "EfficientViT: Lightweight multi-scale attention for high-resolution dense prediction," in Proc. IEEE/CVF ICCV, 2023.
[3] M. Patel, J. Frey, M. Mittal, F. Yang, A. Hansson, A. Bar, C. Cadena, and M. Hutter, "DeFM: Learning foundation representations from depth for robotics," arXiv:2601.18923, 2026.
[4] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret, "Transformers are RNNs: Fast autoregressive transformers with linear attention," in Proc. ICML, 2020.
[5] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus, "Indoor segmentation and support inference from RGBD images," in Proc. ECCV, 2012.